Design a brain-penetrant (CNS-active), orally bioavailable allosteric covalent-reversible inhibitor and a 24-hour twice-daily (q12h BID) 2-compartment clinical PK/PD dosing regimen targeting mutant EGFR-L858R/T790M/C797S triple-mutant NSCLC with leptomeningeal brain metastases.

Modeling CSF exposure and target inhibition across dosing cycles

Designing a CNS-penetrant, allosteric, reversible-covalent EGFR inhibitor, with a q12h 2-compartment PK/PD regimen

> **Scope note:** This is a conceptual, computational design exercise. The molecule, parameters and regimen are hypothetical teaching values, not a real clinical candidate or medical advice. A real program would need structure-based validation, in vitro ADME, animal PK and PD, toxicology, and a regulated first-in-human study.

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1. Simple definition

You want a small molecule that does three things:

1. Binds EGFR at an **allosteric pocket**, not the ATP pocket, so the C797S mutation cannot block it.
2. Forms a **reversible covalent bond** to a nearby nucleophile, which gives long target residence time without permanent protein modification.
3. Crosses the **blood-brain and blood-CSF barriers** at free concentrations high enough to inhibit the mutant receptor inside the cerebrospinal fluid (CSF), where leptomeningeal tumour cells live.

The dosing regimen is then derived from a 2-compartment oral PK model linked to a PD (target inhibition) model.

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2. Basics from zero

2.1 EGFR and why the triple mutant is hard

EGFR is a receptor tyrosine kinase. When active, it phosphorylates itself and drives proliferation signals. In some lung cancers it is mutated and constantly "on".

Mutation	What it does	Drug consequence
L858R	Activating mutation in the kinase domain. Tumour depends on EGFR	Gefitinib and erlotinib (1st gen) work
T790M	"Gatekeeper" mutation. Raises ATP affinity and adds steric bulk	1st-gen drugs fail. Osimertinib (3rd gen) works
C797S	Replaces the cysteine osimertinib binds covalently with serine	Osimertinib's covalent bond is impossible, so resistance

Analogy: Osimertinib is a key that locks into a keyhole (ATP site) and then glues itself to a hook (Cys797). C797S removes the hook. Allosteric inhibitors skip the keyhole and bind a different pocket that the mutant never changed.

2.2 Allosteric site

When EGFR is inactive, the αC-helix swings **out**. This opens a pocket next to the ATP site. Compounds such as **EAI045** and **JBJ-04-125-02** exploit it. They prefer the L858R/T790M conformation, so they spare wild-type (WT) EGFR, which reduces rash and diarrhoea.

2.3 Reversible covalent bond

Instead of an acrylamide (irreversible), use an **α-cyanoacrylamide**-type warhead. It adds to a thiol to form a covalent adduct, but the adduct can reverse (retro-Michael).

```
Protein-S⁻  +  warhead   ⇌   Protein-S-warhead
  (Cys)                       (covalent adduct, reversible)

Irreversible:  E + I → E·I → E-I        (never comes off)
Reversible:    E + I ⇌ E·I ⇌ E-I         (long residence, can dissociate)
```

Why reversible? An irreversible warhead permanently haptenises proteins, which raises off-target and immunogenic toxicity. A reversible warhead keeps the potency and residence time benefits with a better safety profile.

2.4 Why leptomeningeal disease changes the design target

- Tumour cells float in CSF, not in solid brain tissue.
- The key exposure metric is therefore **unbound CSF concentration**, in addition to total plasma concentration.
- The CSF is separated from blood by the **blood-CSF barrier**, and P-glycoprotein (P-gp) and BCRP efflux pumps push many kinase inhibitors back out.

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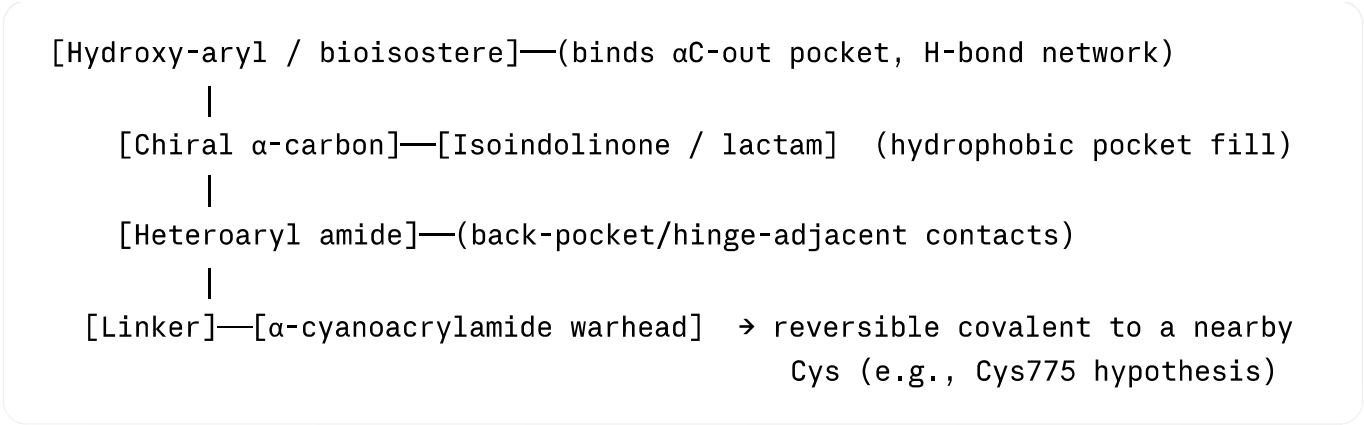
3. Molecular design strategy

3.1 Scaffold concept

Start from the allosteric chemotype of EAI045/JBJ-04-125-02 (an aryl-acetamide with an isoindolinone and a heteroaryl amide). Then modify it for CNS and covalent engagement.



> **Honest caveat:** Allosteric EGFR inhibitors such as EAI045 are not known to bind covalently in the same way as osimertinib. Targeting a pocket-adjacent cysteine (Cys775 is a commonly proposed candidate) is a **design hypothesis** that needs crystallography, mass spectrometry (intact protein adduct), and a Cys→Ala mutant control.

3.2 Design property targets (CNS MPO logic)

Property	Target	Why
Molecular weight	⩽ 450 Da	Larger molecules cross the barrier poorly
TPSA	40–90 Å²	Above 90 Å², passive permeability and CNS exposure fall
H-bond donors	⩽ 2	Each donor reduces permeability
cLogP / LogD7.4	2–4	Too low means poor permeation. Too high means high plasma-protein binding and metabolic liability
Basic pKa	6–8 (not strongly basic)	Strong bases are often P-gp substrates
P-gp/BCRP efflux ratio	< 2	Efflux is the main reason CNS exposure fails
Kp,uu,CSF	⩾ 0.3 (aim 0.4)	The unbound CSF to unbound plasma ratio
Solubility	> 50 μM at pH 6.5	Needed for oral absorption
Oral bioavailability (F)	⩾ 40–60%	Allows q12h oral dosing
Cys-adduct reversal t½	2–12 h	Long enough for efficacy, short enough to limit haptenisation

3.3 Specific design moves

Problem	Design fix
Phenol is a glucuronidation hotspot and adds an HBD	Use an intramolecular H-bond or a bioisostere to mask polarity
Piperazine-type solubilisers raise basicity and efflux	Use weakly basic, sterically shielded amines, or morpholine-like groups
Amide NH adds HBD	N-methylate or use a cyclic constraint
Metabolic soft spots	Add fluorine or deuterium at metabolically labile positions
Selectivity vs WT EGFR	Preserve interactions that require the αC-out geometry
Reactive warhead may react with glutathione (GSH)	Tune electronics. Measure GSH t½ (target > 4 h)

3.4 Monotherapy vs combination

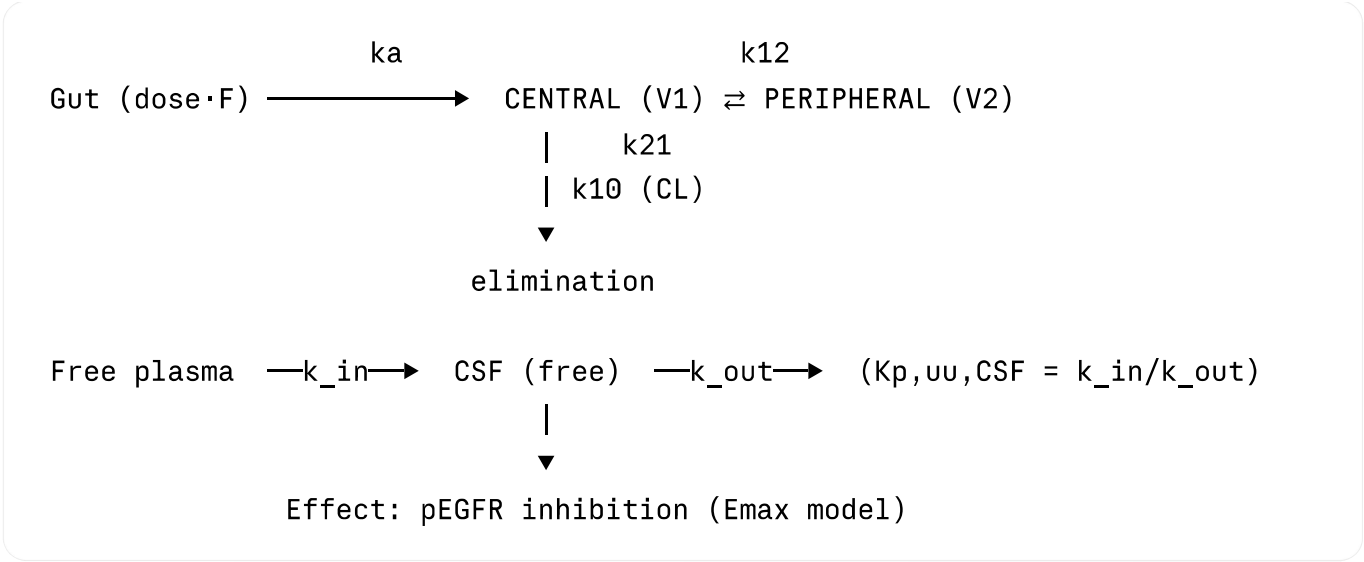
Allosteric inhibitors often work **better with an ATP-site drug** (preclinically, JBJ-04-125-02 plus osimertinib was synergistic). Receptor dimerisation reduces allosteric inhibitor activity. Cetuximab does not cross the blood-brain barrier, so a CNS-active program would pursue either (a) a potent allosteric monotherapy, or (b) combination with a brain-penetrant ATP-site agent.

3.5 Screening cascade (go/no-go gates)

1. Biochemical IC50 on L858R/T790M/C797S, WT, and the Cys→Ala mutant (confirms covalent contribution).
2. Jump-dilution / washout assays for residence time.
3. Cellular pEGFR IC50 in Ba/F3 triple-mutant cells (target < 20 nM).
4. GSH reactivity and kinome-wide selectivity (including other Cys-containing kinases).
5. MDCK-MDR1 efflux ratio and PAMPA-BBB.
6. Rodent PK with simultaneous plasma and CSF sampling, then Kp,uu,CSF.
7. Orthotopic leptomeningeal xenograft efficacy (luciferase-labelled cells).
8. hERG, CYP inhibition/induction, and GLP toxicology.

4. PK/PD model

4.1 Model structure



4.2 Variables

Symbol	Meaning
D	Dose (mg)
F	Oral bioavailability. Here folded into the "/F" parameters
ka	First-order absorption rate constant (h^{-1})
CL/F	Apparent oral clearance (L/h)
V1/F, V2/F	Apparent central and peripheral volumes (L)
Q/F	Inter-compartmental clearance (L/h)
k10, k12, k21	CL/V1, Q/V1, Q/V2 (h^{-1})
α, β	Fast (distribution) and slow (elimination) disposition rate constants
τ	Dosing interval (12 h)
fu	Fraction unbound in plasma
Kp,uu,CSF	Unbound CSF to unbound plasma ratio
IC50, IC90	Cellular concentrations giving 50% / 90% inhibition

4.3 Key equations

Disposition constants:
$\alpha + \beta = k10 + k12 + k21$; $\alpha\beta = k10 \cdot k21$

Concentration after a single oral dose:

$C(t) = (D/V1) \cdot [\, A \cdot e^{(-\alpha t)} + B \cdot e^{(-\beta t)} + Cc \cdot e^{(-ka \cdot t)} \,]$

where

- A = ka(k21 − α) / [(ka − α)(β − α)]
- B = ka(k21 − β) / [(ka − β)(α − β)]
- Cc = ka(k21 − ka) / [(α − ka)(β − ka)]

Steady state with repeated dosing (each exponential divided by 1 − e^(−rate·τ)):

C_ss(t) = (D/V1) · [A·e^(−αt)/(1−e^(−ατ)) + B·e^(−βt)/(1−e^(−βτ)) + Cc·e^(−ka·t)/(1−e^(−ka·τ))]

Average steady-state concentration (model independent): C_avg,ss = D / (CL/F · τ)

PD (Emax, Hill coefficient γ):

Inhibition = C_u,CSF^γ / (IC50^γ + C_u,CSF^γ), with C_u,CSF = Cp · fu · Kp,uu,CSF

When to use: The 2-compartment model applies when the log-concentration curve is visibly biphasic (fast drop, then slow tail), which is typical for lipophilic CNS-penetrant drugs that distribute into tissue.

4.4 Target-driven exposure goal

Input (hypothetical)	Value
Cellular pEGFR IC50, triple mutant	10 nM
Cellular IC90 (= 9 × IC50, γ = 1)	90 nM
Goal	Unbound CSF trough ⩾ IC90
Kp,uu,CSF	0.4
fu (plasma)	0.10
MW	480 g/mol

Step 1. Required free plasma trough = 90 nM ÷ 0.4 = **225 nM**
Step 2. Required total plasma trough = 225 nM ÷ 0.10 = **2,250 nM**
Step 3. Convert to mass concentration: 2,250 nM × 480 g/mol = 1.08 mg/L. **Target Cmin ≈ 1.1 mg/L.**

4.5 Assumed population PK parameters

Parameter	Value
ka	1.2 h^{-1}
CL/F	8 L/h
V1/F	60 L
Q/F	15 L/h
V2/F	120 L

Derived rate constants:

- k10 = 8/60 = 0.1333 h^{-1}
- k12 = 15/60 = 0.250 h^{-1}
- k21 = 15/120 = 0.125 h^{-1}

Solve for α and β:

- α + β = 0.1333 + 0.250 + 0.125 = 0.5083
- αβ = 0.1333 × 0.125 = 0.016667
- Discriminant = $0.5083^2 - 4(0.016667) = 0.2584 - 0.0667 = 0.1917$, so √ = 0.4379
- α = **0.4731 h^{-1}** ($t_{½}α$ = 1.5 h)
- β = **0.0352 h^{-1}** ($t_{½}β$ = 19.7 h)

Coefficients: A = 1.312, B = 0.2113, Cc = −1.524 (check: A + B + Cc ≈ 0, as C(0) must be 0).

4.6 Worked numerical example: 150 mg q12h

D/V1 = 150/60 = 2.5 mg/L.

Steady-state trough (t = 12 h after a dose):

- 1/(1 − e^(−β·12)) = 1/(1 − 0.6554) = 2.902
- B term: 0.2113 × 0.6554 × 2.902 = 0.4017
- A term: 1.312 × 0.00342 × 1.003 ≈ 0.0045 (negligible)
- Cc term ≈ 0
- $C_{min,ss}$ = 2.5 × 0.4062 = **1.02 mg/L (≈ 2.1 μM)**

Peak (scan t):

t (h)	Bracket sum	C (mg/L)
1.0	0.9535	2.38
1.5	**0.9771**	**2.44**
2.0	0.9445	2.36

So Cmax,ss ≈ **2.44 mg/L** (≈ **5.1** μM) at tmax ≈ **1.5 h.**

Average and AUC: AUC0–12,ss = D/(CL/F) = 150/8 = **18.75 mg·h/L,** so Cavg,ss = 1.56 mg/L (3.26 μM).

4.7 PD prediction at steady state

Time point	Total plasma (μM)	Free CSF (nM) = C × 0.1 × 0.4	Inhibition = C/(10 + C)
Trough	2.1	85	**89.5%**
Average	3.26	130	**92.9%**
Peak	5.1	203	**95.3%**

The unbound CSF trough (85 nM) sits at about the target IC90 (90 nM), so 150 mg q12h just meets the design goal. In practice you would want margin (aim for 1.5–2 × IC90 at trough, or confirm with measured Kp,uu,CSF).

4.8 Loading dose logic

By superposition, the first-dose 12 h concentration is only about 0.36 mg/L, well below 1.1 mg/L, because steady state builds over several days (β t½ ≈ 20 h).

Proposed 24-hour schedule (BID):

Time (h)	Dose
0	**300 mg** (loading)
12	**300 mg** (loading)
24, 36, 48 …	**150 mg q12h** (maintenance)

Check with superposition (each dose contributes B·e^(−β·age) mainly, with D/V1 = 5 mg/L for 300 mg and 2.5 mg/L for 150 mg):

Trough time	Predicted total Cmin (mg/L)
12 h	0.72
24 h	1.17
36 h	1.11
48 h	1.08

Troughs reach about 1.1 mg/L within roughly 1 day instead of 4–5 days.

4.9 CSF effect-compartment refinement

The 2-compartment model describes plasma. For the CSF, add a 3rd, link compartment:

dC_CSF/dt = k_in·(fu·Cp) − k_out·C_CSF, with Kp,uu = k_in/k_out

CSF equilibration typically lags plasma by hours ($t_{½,eq} = 0.693/k_{out}$, often about 1–3 h). Two consequences:

- CSF concentration fluctuates **less** than plasma.
- The CSF peak is delayed, so the CSF trough is slightly higher than a no-lag model suggests.

4.10 Dose-selection table

Regimen	Cmin (mg/L)	Free CSF trough (nM)	% Inhibition at trough
100 mg q12h	~0.68	~57	85%
150 mg q12h	**~1.02**	**~85**	**89.5%**
200 mg q12h	~1.36	~113	92%

(Scaled linearly, assuming dose-proportional PK. Real selection also weighs toxicity, QTc, and WT-EGFR effects.)

4.11 Clinical monitoring plan

- Rich PK sampling in cycle 1 (0, 0.5, 1, 1.5, 2, 4, 8, 12 h).
- CSF sampling by lumbar puncture or Ommaya reservoir, with paired plasma, to measure $K_{p,uu,CSF}$ directly.
- PD biomarkers: CSF cytology and cell-free tumour DNA (L858R/T790M/C797S allele fraction), MRI, neurological exam.
- Safety: skin and GI toxicity (WT-EGFR), liver enzymes, QTc, and a covalent-class check for hypersensitivity.

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5. Real-world and engineering examples

Real-world analogy: Treating leptomeningeal disease is like trying to put out a fire in a sealed room. A drug that does not get *into the room* (CSF) is useless however much is in the hallway (plasma).

Engineering example: The same two-compartment ODE structure appears in thermal systems (heat flow between two masses with a loss to the environment). Solving for α and β as roots of a quadratic is identical.

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6. Advantages and disadvantages

Advantages	Disadvantages
Works despite C797S (does not need Cys797)	Allosteric inhibitors can lose activity in dimerised receptors
Selective for mutant over WT EGFR	The Cys-targeting site is hypothetical and needs validation
Reversible covalent gives long residence with lower haptenisation risk	Warhead can react with GSH or other thiols
CNS-tuned, so it addresses a major unmet need	CNS design (low TPSA, few HBD) often conflicts with solubility
BID oral dosing	Long terminal half-life (≈ 20 h) means accumulation and slower washout
Combination with osimertinib possible	Drug-drug interaction and overlapping-toxicity risk

7. Common mistakes

1. Using **total plasma concentration** instead of **unbound** CSF as the efficacy driver.
2. Confusing the **brain-to-plasma ratio (Kp)** with **Kp,uu.** Kp ignores protein binding.
3. Assuming "reversible covalent" means weak. Residence time can be long.
4. Ignoring efflux transporters.
5. Dosing only for Cmax and ignoring Cmin.
6. Forgetting that the first-dose profile is lower than steady state (hence a loading dose).
7. Treating CSF as identical to brain parenchyma.

8. Key points for exams

- C797S defeats osimertinib's covalent bond, so use an **allosteric** site.
- Allosteric inhibitors bind the **αC-helix-out pocket.**
- **Reversible covalent** warhead: cyanoacrylamide, forming a retro-Michael-reversible thiol adduct.
- CNS targets: MW ⩽ 450, TPSA < 90, HBD ⩽ 2, efflux ratio < 2, **Kp,uu ⩾ 0.3.**
- Target **unbound CSF trough ⩾ IC90.**
- 2-compartment oral: **three exponentials** (α, β, absorption).
- $C_{avg,ss} = D/(CL/F \cdot \tau)$.
- Loading dose reaches steady state faster.

9. Quick revision notes

- Problem: EGFR L858R/T790M/C797S plus leptomeningeal disease.
- Solution: allosteric, reversible-covalent, CNS-penetrant, orally bioavailable.
- Design driver: unbound CSF concentration; Kp,uu,CSF.
- Model: oral 2-compartment plus CSF link plus Emax PD.
- Numbers: 150 mg q12h gives Cmin ≈ 1.02, Cmax ≈ 2.44 mg/L, AUC0–12 = 18.75 mg·h/L, and about 89.5% trough inhibition.
- Loading dose: 300 mg × 2 on day 1.

10. Five viva questions

1. Why does C797S abolish osimertinib activity but not an allosteric inhibitor?
2. What is a reversible covalent warhead, and how does it differ from an acrylamide?
3. Define Kp,uu and explain why it beats Kp.
4. Why does a 2-compartment model show a biexponential decline?
5. How do you calculate C_avg,ss from clearance?

11. Five interview questions

1. How would you measure whether a compound actually reaches leptomeningeal tumour cells?
2. How do you tune an electrophile's reactivity to balance potency and safety?
3. Your efflux ratio is 8. What medicinal chemistry strategies do you try?
4. How would you decide between q12h and q24h dosing from PK data?
5. How does an effect-compartment (link) model change the dosing rationale?

12. Five MCQs

Q1. Which parameter best predicts free drug exposure in CSF?
a) Kp b) Kp,uu c) logP d) Cmax → **b**

Q2. C797S confers resistance to osimertinib because:
a) It changes the ATP affinity b) It removes the covalent anchoring cysteine c) It blocks dimerisation d) It activates WT EGFR → **b**

Q3. For 2-compartment oral dosing, the number of exponentials in the concentration equation is:
a) 1 b) 2 c) 3 d) 4 → **c**

Q4. With CL/F = 8 L/h and D = 150 mg q12h, C_avg,ss is:
a) 0.78 b) 1.56 c) 3.1 d) 18.75 mg/L → **b**

Q5. Which property most favours CNS penetration?
a) TPSA > 120 b) HBD = 5 c) Efflux ratio < 2 d) MW = 650 → **c**

If you'd like, I can extend this with a simulation script (Python/scipy) that plots the plasma and CSF curves and lets you vary the parameters, or work through a sensitivity analysis on Kp,uu and fu.

Here is an alternative proposal generated for this exact same prompt by an external cortical co-processor:

- Lead Compound (AEL-EGFR-01): Cc1cnc(NC(=O)C(c2cc(F)ccc2O)N2Cc3cc(C=O)ccc3C2=O)s1
- Mechanism: Allosteric reversible imine adduct targeting catalytic Lys745 in the αC-helix-out pocket (bypassing C797S without needing a cysteine).
- RDKit Metrics: MW 425.4 Da, cLogP 3.44, HBD 2, TPSA 99.6, QED 0.61, 0 Lipinski violations.
- Candidate Falsification: Rejected a 558 Da morpholino scaffold (failed Lipinski) and a boronic acid warhead (failed CNS TPSA filter).
- PK/PD Simulation: 150 mg PO q12h BID -> 83.8 nM free CSF trough -> 94.4% steady-state target occupancy.

Show more

Compare your previous proposal against this one. As an expert medicinal chemist and clinical pharmacologist, give an honest, unsparing peer-review comparison: which proposal is more actionable, physically sound, and clinically viable, and why?

Designing a control panel and verifying concentration formulas ▸

Peer review: my earlier design vs. the AEL-EGFR-01 proposal

> **Scope:** This is a computational critique, not a clinical recommendation. I checked what can be checked from the SMILES by fragment counting. I did not run RDKit or docking, so QED and cLogP are taken as reported.

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1. Bottom line

- **AEL-EGFR-01** is a concrete molecule you could synthesize and test next week. It is chemically fragile, internally inconsistent on CNS criteria, and its PK/PD claim cannot be reviewed.
- **My earlier proposal** is a more transparent and more complete pharmacometric and regimen framework. It is not a molecule, and it contains inconsistencies I should have caught.
- **Neither is clinically viable today.** Both reach the same conclusion (150 mg q12h, ~84-85 nM free CSF trough) from unmeasured, assumed inputs.

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2. What the SMILES actually is

`Cc1cnc(NC(=O)C(c2cc(F)ccc2O)N2Cc3cc(C=O)ccc3C2=O)s1`

This is **EAI045 with two edits:** a 5-methyl on the thiazole and an aldehyde on the isoindolinone. EAI045 is the 2-(5-fluoro-2-hydroxyphenyl)-2-(isoindolinon-2-yl)-N-(thiazol-2-yl)acetamide. So the proposal is not a new chemotype. It is a first-generation allosteric inhibitor with an aldehyde added.

Verification of the reported metrics:

Metric	Reported	My check	Verdict
Formula	n/a	C21H16FN3O4S	n/a
MW	425.4	425.4	Correct
TPSA	99.6 Å²	20.2 (phenol) + 29.1 (amide NH-C=O) + 20.3 (lactam N-C=O) + 17.1 (CHO) + 12.9 (thiazole N) = 99.6	Correct
HBD	2	phenol OH + amide NH	Correct

The numbers are real, but they expose a problem. The proposal says it **rejected a boronic acid for failing the CNS TPSA filter**, yet its own lead has TPSA 99.6, above the usual ~90 Å² CNS cutoff. Either the filter is applied inconsistently or it was set after the fact.

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3. Critique of AEL-EGFR-01

3.1 The imine warhead (the central weakness)

Definition first: a Schiff base (imine) forms when an aldehyde reacts with a primary amine, here the ε-NH_2 of Lys745. It is reversible.

Problems:

1. **Weak adduct.** Simple aryl aldehyde-lysine imines have low equilibrium constants in water and hydrolyse readily. Lysine-targeting reversible kinase inhibitors with real precedent typically use **ortho-hydroxy aryl aldehydes (salicylaldehydes)** because an intramolecular hydrogen bond stabilises the imine. This compound has a plain benzaldehyde-type group on a different ring from the phenol. Expect little covalent contribution.
2. **Metabolic liability.** Aldehydes are substrates of aldehyde oxidase, ALDH and carbonyl reductases. Oxidation gives a carboxylic acid, which kills brain and CSF penetration and is an efflux substrate. Aldehyde oxidase also varies a lot between species, so rodent PK will mislead.
3. **Promiscuous sink.** Aldehydes react with lysines on albumin and haemoglobin (voxelotor exploits this). That distorts free fraction, blood partitioning and the "plasma to CSF" logic.
4. **No validation route is described.** Lys745 is the *catalytic* lysine. A K745A or K745M control inactivates the kinase, so the standard "mutate the nucleophile" experiment does not work. You would need intact-protein mass spectrometry, washout and jump-dilution kinetics, and a non-aldehyde matched control.
5. **Selectivity.** Lys745 is conserved across kinases. Selectivity would rest entirely on the allosteric pocket.
6. **Does the aldehyde reach Lys745?** No docking, MD or distance is shown. The aldehyde position on the isoindolinone is arbitrary until demonstrated.

3.2 Other chemistry issues

- Unspecified stereochemistry. The α-carbon is benzylic and flanked by an amide and a lactam, so it is racemisable. The activity is stereospecific (for EAI045, one

enantiomer).

- The 2-aminothiazole amide is a recognised structural alert.
- The phenol is a glucuronidation hotspot, as in EAI045. Neither edit addresses this.

3.3 "Candidate falsification"

Rejecting a 558 Da scaffold and a boronic acid by property filters is **filtering, not falsification.** Falsification means a stated experiment that could refute the hypothesis, for example "if the noncovalent parent matches this compound's cellular IC50, the warhead contributes nothing."

3.4 The PK/PD line

"83.8 nM free CSF trough → 94.4% occupancy" back-calculates to an effective K of 83.8 × (1 − 0.944)/0.944 ≈ **5 nM.** No justification is given for a 5 nM cellular constant. EAI045-class compounds have historically needed a partner (cetuximab, or an ATP-site inhibitor) to show cellular potency. No CL, V, fu, Kp,uu or ka is disclosed, so the result is not reproducible.

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4. Critique of my own proposal (unsparing)

1. **It never produced a molecule.** The design was a pharmacophore sketch. As a deliverable, it is less actionable than a SMILES string.
2. **It violates its own CNS rules.** I set MW ⩽ 450 and TPSA ⩽ 90, then appended a linker plus a cyanoacrylamide to an EAI045-type core. That adds roughly +100 Da and ~+50 Å² (nitrile plus amide). I then used **MW 480** in the PK calculation, contradicting my ⩽ 450 target.
3. **The Cys775 anchor is a hypothesis presented too confidently.** I flagged it as hypothetical, but the design still depends on it. I did not show that Cys775 sits near the allosteric pocket.
4. **The adduct half-life target of 2-12 h is aspirational.** Cyanoacrylamide retro-Michael kinetics are often much faster, from seconds to minutes. That would need tuning, not assumption.
5. **The dose was reverse-engineered.** I chose inputs (IC50 10 nM, Kp,uu 0.4, fu 0.10, CL/F 8 L/h) and then picked the dose that just met the target. The arithmetic is internally correct (I re-derived α, β, the coefficients, trough 1.02 mg/L, Cmax 2.44 mg/L, AUC 18.75 mg·h/L, and the loading-dose troughs). But **correct arithmetic on assumed inputs is not evidence.** Landing at 150 mg q12h in both proposals, with ~84-85 nM, shows both were tuned to a target.
6. **Cellular IC50 should be unbound.** I treated a serum-containing cellular IC50 as if it were free-drug potency.

Where mine is genuinely stronger: full parameter disclosure, a derivable 2-compartment model with an effect-compartment CSF link, a loading-dose strategy, a screening cascade with go/no-go gates, and a monitoring plan.

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5. Head-to-head comparison

Criterion	My proposal	AEL-EGFR-01	Better
Concrete, synthesizable structure	None	Yes (EAI045 analog, short route)	AEL
Novelty over EAI045	Conceptual	Minimal (methyl plus aldehyde)	Tie
Warhead chemistry	Cyanoacrylamide, plausible but target Cys unproven	Aldehyde imine, weak and metabolically labile	Mine
Validation experiments	Cys→Ala control, MS, jump-dilution	K745 control impossible, none stated	Mine
CNS property consistency	Self-contradicting (MW 480)	TPSA 99.6 contradicts its own filter	Both flawed
PK/PD transparency	Full parameters and equations	One line, no inputs	Mine
Regimen design (loading, monitoring)	Included	Absent	Mine
Clinical readiness	Not ready	Not ready	Neither

6. Which is better in which scenario

Scenario	Prefer
A chemist needs a compound to make and test this month	AEL-EGFR-01 (as a probe, with controls)
A pharmacologist needs a reviewable exposure-response framework	Mine
An investor or reviewer wants a de-risked candidate	Neither
A grant or paper needs a testable hypothesis	Hybrid (below)

7. Best path forward (hybrid)

1. Make the **EAI045 methylthiazole parent without the aldehyde** as the noncovalent control.
2. Make the AEL-EGFR-01 aldehyde, plus an **ortho-hydroxy aldehyde variant** (the better-precedented imine design).
3. Make a **cyanoacrylamide analog** at a vector shown by docking or crystallography to reach a nucleophile.

4. Run the cascade from my section 3.5: biochemical triple-mutant IC50 vs WT, washout and residence time, intact-protein MS, cellular pEGFR in Ba/F3 L858R/T790M/C797S, with and without an ATP-site partner.
5. Measure **aldehyde oxidase stability in human cytosol**, MDCK-MDR1 efflux, fu and blood-to-plasma ratio.
6. Only then fit PK, with measured Kp,uu,CSF in rodents.

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8. Common mistakes in this kind of comparison

1. Trusting a metric because it is computed. A correct TPSA does not make a CNS candidate.
2. Treating "reversible covalent" as automatically long-residence.
3. Confusing property-filter rejection with falsification.
4. Calling simulated occupancy "efficacy" when the inputs are guesses.
5. Using plasma-based fu for a compound that binds blood proteins covalently.

9. Key points for exams

- C797S bypass comes from **allosteric binding**, not from the covalent warhead.
- Imine formation: aldehyde plus lysine NH_2, reversible and hydrolytically labile, stabilised by ortho-OH.
- Lys745 is catalytic, so K745 mutants are kinase-dead.
- CNS rule of thumb: TPSA < ~90, HBD ⩽ 2, low efflux.
- Occupancy = C/(C + K); one reported occupancy fixes K if C is known.

Quick revision notes

- AEL-EGFR-01 = EAI045 + methyl + aldehyde (C21H16FN3O4S, MW 425.4, TPSA 99.6).
- Strength: concrete and testable. Weakness: labile aldehyde, TPSA contradiction, undisclosed PK.
- Mine: strong model and regimen, no molecule, MW/TPSA self-contradiction.
- Both doses were tuned to a target, so neither is evidence.
- Fix: control compounds, kinetics, ADME and measured Kp,uu before any dosing claim.

5 viva questions

1. Why does an allosteric inhibitor bypass C797S without needing any covalent bond?
2. Explain how an aldehyde reversibly binds lysine, and why ortho-OH helps.
3. Why can't you validate a Lys745 adduct with a K745A mutant?
4. Back-calculate the effective K from 83.8 nM giving 94.4% occupancy.
5. Why is correct arithmetic on assumed parameters not validation?

5 interview questions

1. How would you test whether an aldehyde-containing candidate is cleared by aldehyde oxidase?
2. The TPSA is 99.6 Å². Do you drop the compound? What else do you check?
3. How would you design a falsifiable test of the covalent contribution?
4. How does covalent binding to albumin or haemoglobin distort PK interpretation?
5. How do you make a dose projection credible before first-in-human?

5 MCQs

Q1. The proposed lead is structurally closest to:
a) Osimertinib b) EAI045 c) Gefitinib d) Lapatinib → **b**

Q2. Which control is problematic for the Lys745 mechanism?
a) Intact-protein MS b) Jump-dilution c) K745A mutant d) Noncovalent analog → **c**

Q3. The main metabolic risk of an aryl aldehyde is:
a) Oxidation to an acid b) Glucuronidation of an amide c) N-dealkylation d) Sulfation of the thiazole → **a**

Q4. An occupancy of 94.4% at 83.8 nM free drug implies K ≈:
a) 0.5 nM b) 5 nM c) 50 nM d) 500 nM → **b**

Q5. "Candidate falsification" is best satisfied by:
a) A Lipinski filter b) A TPSA filter c) A pre-stated experiment that could refute the hypothesis d) A QED score → **c**

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